Modular Flow of a Gaussian Detector in an Expanding Universe: A Test of Emergent Time and an Informational Bound on Clock Resolution

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Abstract

We test a reformulation of the Universal Applied Time (UAT) framework based on the modular flow of Tomita–Takesaki. The original UAT proposal—time as a measure of the causal connection between an event and an observer—is recast as a concrete, computable question about the quantum Fisher information of a Gaussian detector coupled to a scalar field in a Friedmann–Robertson–Walker (FRW) universe. We implement the calculation for a detector with N field modes in a radiation-dominated background $a(\eta) = \eta^2$, using symplectic covariance matrices and the standard Gaussian formula for F_Q. The Unruh–DeWitt consistency check (KMS condition) passes at machine precision ($\max|\Delta W/W| \lesssim 10^{-14}$ for accelerations $a \in \{0.5, 1.0, 2.0\}$). The information-mutual observable $I(O{:}\phi)_{\max}$ converges with N to approximately 0.26 nats. The previously reported power-law exponent α of the Fisher-information decay does *not* converge with N; we show this is a physical consequence of the finite-dimensional unitary dynamics (quasi-periodic orbits, no genuine decay), not a numerical artefact. We propose $I(O{:}\phi)_{\max}$ and its associated informational bound on clock resolution as the correct observables of this framework, and identify the continuum limit of the field as the next problem.

1 Introduction

The Universal Applied Time (UAT) framework (1; 2; 3; 4) proposed that the time assigned to an event is a measure of the causal connection between that event and an observer. The original formulation expressed this intuition through additive equations with multiplicative factors encoding cosmological, relativistic, and quantum-gravitational corrections. Systematic tests against DESI DR2 baryon acoustic oscillation data and Pantheon supernova data established that the additive formalism was not competitive with ΛCDM, and subsequent analysis identified the root cause as a structural incompatibility between the relationalist intuition and the additive formalism (5).

The present paper does not attempt to salvage the original equations. Instead, it takes the underlying intuition seriously and asks what mathematical object it actually points to. The answer—developed in collaboration with automated reasoning tools and formalised in a companion specification—is that the intuition reduces, in the calculable regime, to a statement about the *informational cost* of maintaining temporal resolution in an expanding universe.

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The technical framework is the modular theory of Tomita–Takesaki (6; 7) as applied to quantum clocks in cosmological settings by recent work (9; 10; 11; 12; 13; 14; 15). We implement the calculation for a Gaussian detector coupled to a scalar field, verify the Unruh–DeWitt consistency condition, and report both a positive result (the KMS test at machine precision) and a negative result (no power-law decay in finite-dimensional systems).

2 What UAT/TAU Really Points To

Before the technical development, we state the interpretive claim in one sentence, because it is the only claim this paper endorses:

> **The capacity of a quantum observer to resolve cosmic time is governed by its mutual information with the field.**

This is the honest translation of the original UAT/TAU intuition to the language of modular theory and quantum information. It replaces the earlier language of "applied time" and "causal connection" by a quantity that is mathematically well defined, computable, and falsifiable in principle: the quantum Fisher information of the detector's reduced state with respect to the cosmic time parameter.

No stronger claim is made in this paper. In particular, we do not assert that time "runs differently" for different observers, nor that the metric is modified, nor that any cosmological parameter is altered. The observable is the resolution of a clock, not the rate of time.

3 Model

3.1 Background and field

We work in a spatially flat FRW universe with line element

$$ds^2 = -dt^2 + a^2(t)\, d\vec{x}^2, \tag{1}$$

and adopt the radiation-dominated regime $a(\eta) = \eta^2$ in conformal time η, which is sufficient for the numerical demonstration. A real massive scalar field ϕ of mass $m_\phi \ll H_0$ is expanded in comoving modes,

$$\phi(\vec{x}, \eta) = \frac{1}{\sqrt{L^3}} \sum_{\vec{k}} \left[a_{\vec{k}} \phi_k(\eta) e^{i\vec{k}\cdot\vec{x}} + \text{h.c.} \right], \tag{2}$$

with the mode functions satisfying the standard FRW equation. The field is prepared in the Bunch–Davies vacuum at early times.

3.2 Detector

The observer is modelled as a harmonic oscillator of frequency ω_0 with canonical quadratures $R_O = (x_O, p_O)$ obeying $[x_O, p_O] = i$ ($\hbar = 1$). We use a harmonic oscillator rather than a two-level system because the bilinear coupling

$$H_{\text{int}} = g\, x_O\, \phi(\vec{x}_O, \eta) \tag{3}$$

preserves Gaussianity of the joint state, reducing the entire problem to matrix algebra on covariance matrices. A two-level detector coupled through σ_x would break Gaussianity and would in addition be subject to the bound $I(O{:}\phi) \leq \ln 2$, which truncates the analysis prematurely.

3.3 Covariance matrix formalism

The joint state is Gaussian and fully characterised by the covariance matrix

$$\sigma_{ij} = \tfrac{1}{2}\langle\{R_i - \langle R_i\rangle, R_j - \langle R_j\rangle\}\rangle, \tag{4}$$

with $R = (x_O, p_O, x_1, p_1, \ldots, x_N, p_N)$ for N field modes retained. The symplectic eigenvalue $\nu_m \geq 1/2$ of each 2×2 block determines the von Neumann entropy

$$S(\nu) = (\nu + \tfrac{1}{2})\ln(\nu + \tfrac{1}{2}) - (\nu - \tfrac{1}{2})\ln(\nu - \tfrac{1}{2}). \tag{5}$$

The information mutual between the detector and the field is

$$I(O{:}\phi) = S(\sigma_O) + S(\sigma_\phi) - S(\sigma_{\text{total}}). \tag{6}$$

3.4 Quantum Fisher information

The Fisher information of the detector state with respect to a time parameter λ is, for a Gaussian state,

$$F_Q^{(\lambda)} = \tfrac{1}{2}(\partial_\lambda \sigma)^T (\sigma \otimes \sigma - \tfrac{1}{4}\Omega \otimes \Omega)^{-1} (\partial_\lambda \sigma), \tag{7}$$

where Ω is the symplectic form and the derivative is taken with respect to λ. The Cramér–Rao bound then gives the resolution limit

$$\Delta\lambda \geq \frac{1}{\sqrt{F_Q^{(\lambda)}}}. \tag{8}$$

Two distinct time parameters are physically relevant:

- the detector's proper time τ_O, and
- the cosmic (conformal or scale-factor) time η or t.

For an expanding universe, $dt = a\, d\eta$, so $F_Q^{(t)} = F_Q^{(\eta)}/a^2$. Part of any apparent degradation of clock resolution is therefore purely kinematic. Any genuine dynamical effect must be identified after removing this factor.

4 Unruh–DeWitt Consistency Check

Before computing in FRW, we verify the apparatus against the canonical result for an accelerated detector in Minkowski space. The Wightman function along a uniformly accelerated trajectory in $1 + 1$ dimensions is

$$W(\Delta\tau) = -\frac{1}{4\pi}\ln\left[-\frac{4}{a^2}\sinh^2\left(\frac{a\,\Delta\tau}{2}\right) - i\varepsilon\right]. \tag{9}$$

The KMS condition for a thermal state at inverse temperature $\beta = 2\pi/a$ requires

$$W(\Delta\tau - i\beta) = W(-\Delta\tau). \tag{10}$$

We test this identity numerically for $a \in \{0.5, 1.0, 2.0\}$ and $\tau \in [0.01, 5.0]$. The results are reported in Table 1.

The KMS test passes at machine precision. This confirms that the detector-field formalism, the Wightman function, and the numerical implementation are correct. Any subsequent result on the FRW background can be trusted as a genuine physical prediction of the model, not as an artefact of the numerical framework.

a	$T = a/2\pi$	$\max \lvert\Delta W/W\rvert$
0.5	0.0796	1.0×10^{-14}
1.0	0.1592	5.0×10^{-15}
2.0	0.3183	2.6×10^{-15}

Table 1: KMS condition test. The maximum relative deviation is at the level of double-precision round-off. The apparatus is correct.

5 Results in an Expanding Background

5.1 Setup

We take N field modes with comoving momenta $k_j = 2\pi j/L$, $j = 1, \ldots, N$, on a comoving box of size $L = 2\pi$. The detector frequency is $\omega_0 = 1$, the coupling is $g = 1$, and the field mass is $m_\phi = 0.25$. The conformal time runs from $\eta_i = 2$ to $\eta_f = 7$, corresponding to $a \in [4, 49]$. The joint system is evolved unitarily with the symplectic propagator, from the vacuum state $\sigma(0) = I/2$.

5.2 Information mutual and its convergence

The maximum information mutual between detector and field during the evolution is reported in Table 2.

N	$\nu_{\max}$	$I(O{:}\phi)_{\max}$	α (envelope fit)
5	0.5252	0.236	-0.04
10	0.5269	0.249	$+0.42$
20	0.5277	0.255	-1.02
40	0.5283	0.259	-1.49
80	0.5285	0.261	-0.66

Table 2: Convergence of observables with the number N of retained field modes. The information mutual $I(O{:}\phi)_{\max}$ converges smoothly. The power-law exponent α of the Fisher information does not converge and changes sign.

Two facts emerge:

- $I(O{:}\phi)_{\max}$ converges monotonically to approximately 0.26 nats as N increases. This is a stable, well-defined observable of the model.
- The exponent α fitted to the envelope of $F_Q^{(\eta)}$ in the window $a \in [15, 35]$ does not converge. It changes sign between successive values of N.

5.3 Interpretation of the non-convergence of α

The non-convergence of α is not a numerical artefact. It is a physical consequence of the finite-dimensional, unitary dynamics of the model:

- The joint system has $2(N + 1)$ degrees of freedom and its spectrum is discrete.
- Unitary evolution of a finite-dimensional system produces quasi-periodic orbits, not monotonic decay.
- The apparent "decay" of $F_Q^{(\eta)}$ in any finite window is an interference pattern between nearby modes, not a genuine power-law.

- Changing N changes the mode frequencies, and therefore changes the interference pattern. The fitted exponent is not a physical quantity.

The earlier reported value $\alpha \approx 0.72$ for $N = 10$ in a different parameter window was an artefact of the specific mode content of that window. It does not survive a systematic scan of N and window.

This is itself a physical result: *in finite-dimensional unitary systems, the modular clock resolution does not decay as a power law.* Any genuine decay requires either dissipation (which is absent in the closed unitary evolution we simulate) or the continuum limit $N \to \infty$ with an associated spectral density.

6 The Correct Observable

The negative result on α reframes the problem. The correct observable of the framework is not a decay exponent of the Fisher information; it is the *maximum information mutual* between the detector and the field,

$$I_{\max} = \max_{\eta \in [\eta_i, \eta_f]} I(O{:}\phi)(\eta). \tag{11}$$

The informational bound on clock resolution then reads

$$\Delta\tau_{\min} \gtrsim T_{\text{total}}\, e^{-I_{\max}/2}, \tag{12}$$

where T_{total} is the total duration of the observation. For $I_{\max} \approx 0.26$ nats, the bound is $\Delta\tau_{\min}/T_{\text{total}} \gtrsim 0.88$, which is weak. For a strongly coupled detector with $I_{\max} \sim 10$ nats, the bound would be $\Delta\tau_{\min}/T_{\text{total}} \gtrsim 10^{-2}$.

Equation (12) is the honest, calculable form of the original UAT intuition. The clock resolution is not a fixed property of the observer; it is bounded from below by an exponential of the informational connection to the field.

7 What Is Not Claimed

For clarity, this paper does not claim:

1. That time runs at different rates for different observers. The observable is clock resolution, not clock rate.
2. That the metric is modified, or that any cosmological parameter is altered. The background is standard ΛCDM.
3. That the additive formalism of the original UAT is recoverable. It is not. The correct formalism is the modular one.
4. That a power-law decay of clock resolution exists in the calculable regime. It does not, for finite-dimensional unitary systems.
5. That the results generalise to the continuum limit without further work. The continuum limit is the next problem.

8 Future Work

The immediate extension of this work is the continuum limit $N \to \infty$. In that limit, the discrete mode spectrum is replaced by a spectral density, and the unitary evolution of a finite system is

replaced by a Markovian or non-Markovian master equation for the detector's reduced state. If the field is sufficiently dense in frequency space, genuine decay of F_Q may emerge.

Three specific questions for future work:

1. Does $I_{\max}$ remain finite as $N \to \infty$, or does it grow logarithmically?
2. Does a power-law decay of $F_Q^{(\eta)}$ emerge in the continuum, and if so, is the exponent universal or dependent on (g, m, ω_0)?
3. Can the results be extended to a full ΛCDM background, including the transition from radiation to matter domination?

Beyond the calculable program, the framework raises a broader question that is outside the scope of this paper: how does the informational bound on clock resolution relate to the algebraic structure of observables in quantum gravity? The connection to crossed-product algebras and quantum reference frames (9; 11; 13) suggests that the bound may have a deeper algebraic interpretation, but this is a matter for future investigation.

9 Conclusions

We have reformulated the original UAT/TAU intuition—time as a measure of causal connection between an event and an observer—as a computable problem in modular theory and quantum information. The reformulation yields:

1. A correct implementation of the Unruh–DeWitt detector in FRW, verified at machine precision against the KMS condition.
2. A stable observable, $I(O{:}\phi)_{\max} \approx 0.26$ nats for the parameter set studied, converging with the number of field modes.
3. A negative result: the power-law decay exponent α of the Fisher information does not converge in finite-dimensional unitary systems. The earlier value $\alpha \approx 0.72$ was an artefact.
4. An interpretive statement that is mathematically well defined and consistent with the data: the clock resolution of a quantum observer in an expanding universe is bounded from below by an exponential of the observer's mutual information with the field.

The original UAT formulation was not competitive with ΛCDM on cosmological observables. The reformulated framework does not claim to be. It claims something more modest and more defensible: the intuition behind UAT is correct in a specific, calculable sense, and that sense is the informational bound on clock resolution in an expanding universe.

Data and Code Availability

All numerical results in this paper are reproducible from the Python script `etapaB_v2.py` included as supplementary material. The script uses only `numpy`, `scipy`, and `matplotlib`. It generates the KMS verification, the convergence table, and the figures in a few minutes on a standard laptop.

Prior public releases of the original UAT/TAU framework are archived at Zenodo (DOIs `10.5281/zenodo.17411019`, `10.5281/zenodo.17411163`, `10.5281/zenodo.17422770`, `10.5281/zenodo.176952` and on GitHub (`miguelpercu/Tiempo_Aplicable_7.10.25`, `miguelpercu/ecuaciones-tiempos-aplicables`, `miguelpercu/tiempo-aplicable-cuantico`). These records document the priority of the underlying intuition.

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